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Change management scenario to improve Webometrics ranking

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Abstract

Webometrics ranking of world universities (WRWU) is recognized as a trusted ranking system for the world's universities. Previous study shows that there is a relationship between the electronic Word of Mouth (e-WOM) and Web Content Accessibility Guidelines (WCAG) 2.0 variables on the increase of the webometrics ranking indicator value. This research is improving the previous result by mapping the recommendations into element based on the Awareness, Desire, Knowledge, Ability, Reinforcement (ADKAR) method. The goal of this research is creating recommendation activities and change management scenario in ADKAR method to improve university webometrics ranking. The result of this study is 75 recommended activities to improve university webometrics ranking.

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Keywords: ADKAR; change management; webometrics; WRWU

1. Introduction

Indonesian Government has targeted several universities in Indonesia to enter the ranks of world-class universities (WCU) [1]. One of the references used for world-class university rankings is the Webometrics Ranking of World Universities (WRWU). Webometrics ranking assessment is carried out based on the quality of the page (website) and the vast amount of information the university has on the internet. The website is considered as reliable representation of the quality of higher education institutions, so universities must improve the quality and quantity of website content in accordance with WRWU standards [2]. There are various aspects that must be addressed in the process of increasing webometrics rankings, such as website quality, human resources, finance, etc. University can focus on improving the

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quality of the website as the first step in the process of increasing webometrics rankings. A strategy is needed to improve the quality of the website so that universities can meet the target which is to be at the top of the WRWU. Indicator used for WRWU calculation consist of four indicator which is presence, visibility, transparency (openness), and excellence (scholar). Presence indicator defined as public knowledge shared with, and the data is taken from the result of google search. Visibility indicator defined as web contents impact and counted based on the number of external networks (subnets) linking to the institution's webpages. The source of the data is Ahrefs and Majestic. Transparency indicator defined as top cited researchers. The data is counted based on number of citations from top 210 authors (excl. top 20 outliers) from google scholar profile. Excellence indicator defined as top cited papers with the data is taken from Scimago for 5 years period. For July 2020 edition, WRWU indicator consist of 5% Presence, 50% Visibility, 10% Transparency, and 35% Excellence [2].

This study aims to develop recommendations activities that could improve university webometrics rankings. This study also develop change management scenarios for implementation of webometrics ranking improvement strategy. Previous study is using various method such as WCAG 2.0 , e-WOM, Bayesian network, content analysis, etc to make guideline to improve webometrics ranking. To develop the improvement activities, the researcher is mapping previous study results into the phase of the ADKAR method based on study literature. The mapping process into ADKAR method phase is done to simplify the implementation process. Then the modified result is used as the basis in interview with representatives from the case study university. The interview results and observation on the object university are used to develop change management scenarios to implement recommendation activities. The result of this study is recommendations activities to improve the webometrics ranking and change management scenarios to implement it.

2. Literature Review

2.1. Improving Webometrics Ranking

Efforts to improve the university's webometrics ranking began with the webometrics ranking components. The 4 webometrics ranking indicators guideline on the webometrics website can be used as basis to make improvements. Research shows, the improvement attempt can be carried out on the university website using tools and guides. Study by (Elgharabawy & Ayu, 2011) looked for the effect of WCAG 2.0's suitability with Webometrics ranking on 150 websites. The research was conducted on each website that uses the Total Validator to find errors based on the WCAG 2.0 criteria. The results of the study show that there is positive evidence, although weak between accessibility and visibility indicators from webometrics [3]. Research by (Yusinda,2010) produced recommendations and a proposed design in the form of improvements to aspects that were not optimal at the time of the evaluation using WCAG [4]. (Windriyani, P., et. al., 2014) stated that webometrics does not provide criteria that can be used as references by universities in creating websites that support the assessment. Whereas WCAG 2.0 is a stable technical standard and can be used as a reference. So that the new WCAG 2.0 was chosen as a technical assessment in the evaluation of websites with good accessibility [5].

According to (Damayanti & Subriadi, 2015), e-WOM has a significant effect in improving the image of higher education [6]. This research is investigated further in (Amalia,2017) research by calculate site performance measurements based on webometrics indicators and provides recommendations for improvements based on e-WOM [7]. Research by (Hafidz,2018) combining previous research method such as WCAG 2.0 and e-WOM to make a new one. This research mapped four webometrics indicators with four principles of WCAG 2.0 (Perceivable, Operable, Understandable, and Robust) and four e-WOM indicators (positive self enhancement, economic incentive, helping companies, and seeking advice). The result is a set of recommendations standard to improve the university webometrics ranking in Indonesia[8].

2.2. ADKAR Change Management

To implement the recommendations for improving the webometrics ranking, a change management is needed. The research by (Lawler, 2011) resulted in the steps taken for a successful implementation based on the ADKAR method. The research also discusses the challenges faced in carrying out the ADKAR stages[9]. (Sulistiyani, et al, 2020) made a change management roadmap consisting of three states, namely current, transition, and future. This strategy

formulation also proposes the establishment of KPIs to determine the success of each proposed strategy [10]. Research results from (Goyal and Patwardhan,2018) show that after successful implementation, managers and employees are responsible for maintaining change in their organizations for a long time. If change is not reinforced properly and employees go back to their old ways of carrying out their duties, this can lead to failure of the change process[11]. (Fawzy and Bangun, 2014) classifies recommendations to reduce gaps in change management plans based on organizational aspects, human resources, targets, and external factors [12]. According to (Gilani, et al., 2017), the biggest obstacle to implementation is a lack of resources[13]. Other research, such as (Ali, et al., 2016) and (Ali, et al., 2018) produce strategies that can be used for the mapping process of webometrics ranking improvement recommendations to ADKAR[14],[15]. Research by (Purnomo, 2019) mapping recommendation for improving webometrics ranking from (Hafidz, 2018) research into ADKAR elemens, and the making a set of activities for every strategies [16].

3. Research Method

3.1. Research Object

This research is conducted in UPN Veteran East Java. UPN Veteran East Java was chosen because it meets the research requirements of a university that needs change management to improve their webometrics ranking. For the last 3 years from January 2017 – July 2020, UPN Veteran East Java's webometrics ranking is still outside of the top 50. Change management is proposed to improve UPN Veteran East Java webometrics ranking. In the process, researcher make comparison between UPN Veteran East Java and other three universities. The other three universities are University of Indonesia which ranked number 1 in webometrics ranking for July 2020 edition, UIN Sunan Ampel and Narotama University. UIN Sunan Ampel and Narotama University is chosen because both are from same city, Surabaya as UPN Veteran East Java and have similar level of competition.

3.2. Methodology

In this study, the recommendations to improve webometrics ranking based on WCAG 2.0 and e-WOM from (Hafidz,2018) research will be mapped into phase of ADKAR method. The diagram of the research stages can be viewed on Fig. 1

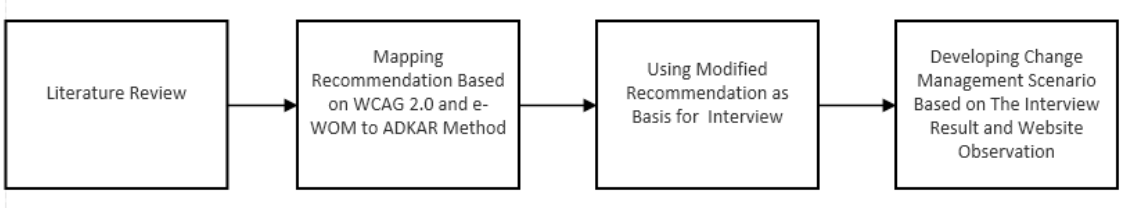


Fig. 1. Diagram of The Research Stages

Before being mapped to the ADKAR method, recommendations were mapped to COBIT 5 at first. Mapping to COBIT 5 was done because ADKAR method activities was general in nature. COBIT 5 has detailed guidance on activities that must be done to support change. (Indrajit, 2013) shows an effective implementation strategy taken from COBIT 5 [25]. From the COBIT 5 implementation life cycle model only the second layer, change enablement, is used for the mapping process [26]. This is because the second layer focuses on empowering change and similar to those of ADKAR change management. The change enablement layer has the same stages as the management practice of the BAI 05 Manage Organizational Change Enablement process from the Build, Acquire, and Implement domains. Mapping is done by looking for similarities between activities of each management practice from COBIT 5 domain BAI 05 with ADKAR activities.

Each recommendations for webometrics improvements from (Hafidz, 2018) research then mapped into each ADKAR change management element based on multiple literature. Each recommendation will be mapped into one or

two element among Awareness, Desire, Knowledge, Ability, and Reinforcement. Table 1 shows example of the mapping process.

Table 1. Example of Mapping Webometrics Recommendations into ADKAR Methods

Indicator	No.	Recommendation	A	D	K	A	R	Reference
Presence	1.	Recommendation 1	√					Reference 1
	2.	Recommendation 2		√				Reference 2
	3.	Recommendation 3			√			Reference 3

The mapping of webometric improvement recommendations based on WCAG 2.0 and e-WOM to the ADKAR model was carried out by looking for similarities between the proposed recommendations and activities from the ADKAR model activities in other literature. The similarity of the statement is chosen accordingly in the form of an explicit statement that has the same sentence structure or implicit statements where the sentence is not the same but has the same meaning or goal.

3.3. Change Management Scenario

Researchers made observations on each four webometrics ranking indicator criteria. The results of these observations were then compared with the results of the webometrics ranking of the University o f Indonesia as number rank. 1 in Indonesia for July 2020 edition. In addition, the results of these observations were also compared with two universities in Surabaya, namely Narotama University and UIN Sunan Ampel Surabaya. This observation is carried out to determine the present value of each webometric indicator of every university.

Researchers made observations on every university official website between November and December 2020. These observations were made to determine the value of each webometric indicator. Webometric has four indicators for ranking, namely Presence, Visibility, Transparency (Openness), Excellence (Scholar) [2]. The value of each of these indicators is obtained from the search engine according to the terms of the webometrics calculation methodology. For Presence indicator, data is collected from google, Visibility indicator data from Ahrefs and Majestic, Transparency indicator data from Webometric transparency ranking, and Excellence indicator data from scopus.

According to Aguillo, et al. (2010), the numbers need to be log normalized first by adding one unit to avoid log (0), the exclusion of max and min values is excluded (and vice versa for median and average) and then changed. to be ordinal: first order (highest value) is one, second is two and so on [23]. Research by (Amalia,2017) calculates the normalization value of each webometric indicator for ITS Surabaya using the following normalization formula as Eq. (1) :

$$N_a = \frac{\log(n_a + 1)}{\log(\max(n_i) + 1)} \quad (1)$$

With Na is normalized value of each indicator, na is value from search engine, and max(ni): the highest na value of all colleges

The next stage is to do interview, at this stage the writer will collaborate with two interviewees from the object university member of the webometrics ranking team and the website administrator. The interview is done by using modified recommendation that already mapped to ADKAR elements to know the current condition of the university and future condition that wants to be achieved. This interview also looking for possible obstacles that will occur if the recommendations for improvement and change management scenarios are implemented

4. Result and Discussion

Mapping recommendations for improving webometrics based on WCAG 2.0 and e-WOM to ADKAR resulting in 75 recommendation activities. This is consist of 11 recommendation activities for the Awareness stage, 7 recommendations activities for the Desire stage, 26 recommendation activities for the Knowledge stage, 20

recommendation activities for the Ability stage, and 11 recommendation activities for the Reinforcement stage. In order to get the a broader and more detailed picture, this research also created input and output document needed for each recommendation, RACI matrix, as well as gap analysis of present and future condition.

4.1 Change Management Scenario

The result of the observation of each university webometrics indicator point is shown in Table 2.

Table 2. Result of The Webometrics Indicator Observation

Ranking	University Name	Presence (5%)	Visibility (50%)	Transparency (10%)	Excellence (35%)
18	Universitas Narotama	0,754	0,753	0,615	0,545
63	Universitas Islam Negeri Sunan Ampel	0,918	0,930	0,565	0,542
83	Universitas Pembangunan Veteran Jawa Timur	0,758	0,724	0,682	0,572
1	Universitas Indonesia	1,000	1,000	1,000	1,000

From the results of the interviews conducted, the informants asked for more detailed recommendations that could be applied practically. Recommendations that are still general in nature will be changed to more detailed recommendations according to the references obtained.Table 3. is showing the interview result which consist of number of recommendation that already fullfiled in current condition.

Table 3. Interview Result

ADKAR Element	Indicator	Number of Recommendations	Number of Fulfilled Recommendations	Percentage
Awareness	*Presence*	11	5	45%
Desire	*Presence*	7	6	86%
Knowledge	*Presence*	13	11	85%
	Visibility	11	6	55%
	Excellence	2	1	50%
Ability	*Presence*	5	4	80%
	Visibility	11	5	45%
	Transparency	2	1	50%
	Excellence	2	1	50%
Reinforcement	*Presence*	2	2	100%
	Visibility	4	1	25%
	Transparency	3	1	33,3%
	Excellence	2	2	100%

Based on the results of interviews that have been conducted, average percentage for fullfiled recommendations in total are 73.7% for presence indicator, 46.2% for visibility indicator, 40% for transparency indicator, 66.7% for excellence indicator. Researcher then made calculation of current webometrics points and the number of percentage increase if the object university is implementing the recommendation activities. Table 4 shows the results of the webometrics point increase calculation scenario.

Table 4. The Webometrics Point Increase Calculation Scenario

Indicator	Full Score 100%	Percentage	Simulation 75%	Percentage	Simulation 50%	Percentage	Simulation 25%	Percentage
Presence (5%)	1.000	24.20%	0.940	18.15%	0.879	12.10%	0.819	6.05%
Visibility (50%)	1.000	27.60%	0.931	20.70%	0.862	13.80%	0.793	6.90%
Transparency (10%)	1.000	31.80%	0.921	23.85%	0.841	15.90%	0.762	7.95%
Excellence (35%)	1.000	42.80%	0.893	32.10%	0.786	21.40%	0.679	10.70%
Total Score	1.000	24.20%	0.917	18.15%	0.834	12.10%	0.751	6.05%

4.2 Recommendations

To ensure this change management scenario is successful, universities need to take these recommendations into account when carrying out the implementation process steps for implementing a change management scenario. The recommendations are as follows:

a) Element of Awareness
- Create a special team focused to improve webometrics rankings.
- Communicating the importance of increasing webometrics rankings through the media owned to the entire university community.
- Top management's commitment to making changes so the implementation can run until finished.

b) Desire Element
- Stakeholders can appoint and motivate representatives of staff, lecturers, and students to support change.
- Include webometrics ranking improvement activities as a mandatory university agenda.
- Hold blog competitions and scientific works for the academic community.

c) Knowledge Element
- Providing training for university and faculty website management staff.
- Preparation of guide material for steps to improve webometrics rankings for the academic community.
- Collaboration with external parties that can support the smooth running implementation process.
- Making guidelines for blog management for the academic community.

d) Ability Element
- Allocation of resources to implement the webometrics ranking improvement activities.
- Trial to optimize the website feature enhancement.
- Creation of rules and procedures as a guide in implementing webometrics ranking improvement.
- Measurable target in the implementation of blog competitions.

e) Elements of Reinforcement
-Policy to provide rewards for parties who play a major role in the successful improvement of webometrics rankings.
- Encouragement from management so that every faculty supports them blog competitions
- Ensure optimal use of Google scholar and repositories.
- Stakeholders also need to provide realistic targets related to successful implementation targets.
- Periodic evaluation based on feedback and assessment of benefits and costs.
- Strengthening the continuity of changes that occur in the organization

4.3 Discussion

The result of mapping process is 75 recommendation activities. By grouping this recommendations based on ADKAR element it is expected that the implementation of each recommendation in the future will be easier to do. Since changes the ADKAR model are broken down into element so that it can be seen where the changes failed and the impact of the rejection. Recommendation by (Hafidz,2018) is still general in nature so it may change when implemented in other universities it need adjustment such as a detailed activities. While research by (Agung,2019)

needs reference to strengthen the mapping process. New recommendations which cater to detail is needed. This new recommendation can be taken from research (Suksida & Santiworarak, 2017), (Purwiantono, 2019), (Wahyuningrum, 2015), and (Rahardja, et al., 2017) which focus on increasing visibility indicators which have a large percentage (50%)[17],[18],[19],[20]. Likewise, research by (Herliyani, et al., 2017) and (Darmasaputra, et al., 2016) focused on non-technical aspects of increasing webometrics rankings by focusing on GUI and content[21],[22]. While research (Yakymenko, et. al., 2020) can be used as a reference for improvement and simulation of the estimation of webometrics ranking calculations [23].

This study has limitations, since the most of the recommendation is based on previous research (Hafidz, 2018) and (Purnomo, 2019) which focused on improving websites with e-WOM, WCAG 2.0 and ADKAR change management activities to improve webometrics rankings. Among 75 recommendation activities, 38 and 26 activities intended to improving presence and visibility indicator. Only 5 recommendations activities is for transparency indicator and 6 recommendations for excellence indicator. There are big gaps between the numbers of recommendation among those four indicators. Both transparency and excellence indicators need more recommendation activities to improve so it can be a better overall scenario that improves all four indicators equally. to fix this problem may require other methods that are more suitable for the criteria for excellence and transparency indicators . The content of this recommendation activities also mostly focused on website related aspect. There are no recommendations that specifically cover other aspects such as human resource, finance, etc. that also contributing to improving webometrics rankings.

Based on the estimated webometrics points, there are large gap of webometrics point between the research object university and the rank number 1 university. The object university needs to focus on increasing points, especially visibility and excellence indicator. The reason is because both indicator have a large percentage of webometrics point, which is 50% and 35%. Between these two indicators, it will be easier to increase the visibility indicator compared to excellence. This is because the indicator of excellence requires a longer stage because it needs to create a culture of researching and writing journals in both English and Indonesian. Meanwhile, the visibility indicator can be improved by good search engine optimization (SEO). By focusing resources on the visibility indicator, webometrics ranking scores can increase in the short term. According to (Yakymenko et. al, 2020) the process of increasing webometrics rankings in a university requires the development of individual strategies in promoting the university [23]. The strategy is to build a webometrics prediction estimation model, and implement the introspection function. Therefore, university needs to develop an individual strategy based on university priorities in the long, mid, and short term. For the implementation process, in order to make the change management scenario go according to plan, a collaboration with all responsible stakeholder is needed. In order to make change management scenarios, there are direct contributions and correlation between the recommendation to improve WRWU, ADKAR change management, and stakeholders involved in change management. The WRWU indicator encourages the making of recommendations to increase WRWU thus, it does not directly contribute to the development of change management scenarios. Fig 2. shows the correlation between WRWU indicators, recommendations, ADKAR , and stakeholder decision that contribute in developing change management scenario.

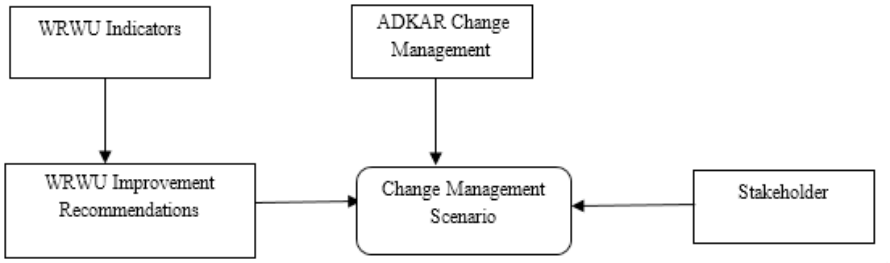


Fig. 2. Correlation Between WRWU Indicators, The Recommendation to Improve WRWU, ADKAR Change Management, and Stakeholders

5. Conclusion and Future Works

5.1 Conclusion

This research goal is to develop recommendation activities and change management scenario in ADKAR method to improve university webometrics ranking. The mapping of webometric improvement recommendations based on WCAG 2.0 and e-WOM to the ADKAR model was carried out by looking for similarities between the proposed recommendations and activities from the ADKAR model activities in other literature. The scenario is based on the results of interviews and observation on the university website. The result of mapping process is 75 recommendation activities in ADKAR element. The change management scenario contains recommended steps for successful implementation of the webometrics ranking improvement strategy.

5.2 Future Research

Future research can implement this proposed recommendations as well as testing it to find out the success rate of this change management scenarios. Further research can complement change management scenarios by adding KPIs (Key Performance Indicators) to measure the achievement success rate. The number of recommendation between all indicator also must be equal and not focusing on one or two indicators only. There are also the needs for more diverse recommendation activites that other aspect such as finance and human resource. The stakeholder and other component of the university also need to be more considered in the process of developing recommendations activities and change management. With the release of webometrics ranking January 2021 edition there are big change in the methodology with presence indicator has been removed from the composition of the webometrics ranking calculation. Future research needs to be upgraded to keep relevant with the new WRWU criteria.

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